

One-Day-Ahead 1% VaR Forecasting on QQQ Daily Log Returns

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1 Introduction

This report studies one-day-ahead forecasting of the 1% Value-at-Risk (VaR) for QQQ daily log returns. The practical question is simple: given information available at the close of day t , how far into the left tail could the next trading day's return fall with 1% probability? In risk management language, the goal is to estimate the conditional 1% quantile of the next-day return distribution and to test whether the resulting forecasts are statistically credible in real time.

The project is implemented as a rolling out-of-sample forecasting exercise using QQQ returns and, for one comparison model, the VIX index as an exogenous volatility signal. Rather than focusing on in-sample fit, the analysis emphasizes a realistic forecasting and backtesting workflow: estimate each model using data available up to time t , produce a one-day-ahead VaR forecast for time $t + 1$, and then record whether the realized return violates that forecast.

The saved project outputs show that empirical-tail GARCH methods perform best in this setting. In particular, GJR-FHS and GARCH-FHS deliver the strongest overall calibration, while Gaussian VaR methods are too thin-tailed for QQQ downside risk. Historical Simulation remains a useful and competitive benchmark. The package-based GARCHX model with VIX is included as a standard exogenous-volatility comparison, but it does not outperform the best empirical-tail models in the saved results.

2 Historical Background

VaR became a dominant market-risk summary in the 1990s as financial institutions needed a single, interpretable measure of downside exposure over a fixed horizon and confidence level. A major turning point was the public release of the *RiskMetrics* framework by J.P. Morgan in 1994, which helped standardize volatility-based risk measurement and made VaR central to both practice and regulation. The Basel market-risk framework in the mid-1990s further accelerated its use by allowing banks to rely on internal VaR systems for regulatory capital calculations.

From a modeling perspective, VaR forecasting is closely tied to the development of conditional heteroskedasticity models. Engle's ARCH model and Bollerslev's GARCH model provided a tractable way to represent volatility clustering, one of the core empirical features of financial returns. In particular, the GARCH(1,1) model became a workhorse benchmark because it allows conditional variance to evolve over time in response to recent shocks and past volatility.

As VaR models became widespread, backtesting became just as important as forecasting. Kupiec (1995) proposed the unconditional coverage test, which checks whether the observed violation frequency matches the nominal VaR level. Christoffersen (1998) extended this logic by adding an independence test, recognizing that a risk model is inadequate if violations cluster in time even

when the overall count looks reasonable. Together, these tests remain standard tools for evaluating VaR forecasts.

A different line of development focused on modeling the quantile directly rather than modeling the full return distribution first. Engle and Manganelli's CAViaR framework formalized this idea by specifying a dynamic process for the VaR itself and estimating it through regression quantiles. This approach is attractive because it avoids strong distributional assumptions, although in practice it can be computationally sensitive and requires careful adequacy checking.

These developments motivate the current project. The main comparison is not just between different volatility recursions, but between different ways of mapping volatility information into a left-tail quantile: Gaussian parametric tails, Student- t tails, empirical residual tails, and direct quantile dynamics.

3 Purpose Statement and Mathematical Development

3.1 Purpose of the topic

The purpose of VaR forecasting is to estimate a conditional lower-tail threshold for returns. For a one-day-ahead VaR at level $\alpha = 0.01$, the ideal forecast satisfies

$$\Pr(r_{t+1} < \text{VaR}_{t+1}(\alpha) \mid \mathcal{F}_t) = \alpha,$$

where r_{t+1} is the next day's return and $\mathcal{F}_t$ denotes the information set available at forecast origin t . In this project, r_t is the daily log return of QQQ,

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right),$$

with P_t denoting the adjusted closing price.

For most conditional volatility models in the project, the return equation takes a location-scale form,

$$r_{t+1} = \mu_{t+1} + \sigma_{t+1} z_{t+1},$$

so that one-day-ahead VaR is computed as

$$\text{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1} q_\alpha,$$

where q_α is either a parametric quantile or an empirical residual quantile.

3.2 Data setting used in this report

The project uses QQQ adjusted close data and VIX close data beginning on 2015-01-01. QQQ prices are converted to daily log returns, while VIX enters only in the GARCHX comparison model as an exogenous volatility regressor. The saved local project outputs indicate:

- modeled sample begins in early 2015;
- a 70% initial training fraction is used in the rolling forecast design;
- the saved out-of-sample backtests contain 848 one-day forecasts per model;
- at the 1% VaR level, the expected number of violations is 8.48.

3.3 Forecasting and backtesting design

At each out-of-sample date, the workflow is:

1. use all observations available through time t as the training sample;
2. estimate the model on that rolling window;
3. produce a one-day-ahead 1% VaR forecast for time $t + 1$;
4. compare the realized return to the forecast and record a violation indicator

$$I_{t+1} = \mathbf{1}(r_{t+1} < \text{VaR}_{t+1});$$

5. evaluate the resulting hit sequence using coverage and independence tests.

This design matters because VaR is a forecasting problem, not merely a descriptive one. A model that fits the historical sample well but fails on rolling out-of-sample evaluation is not useful for risk control.

3.4 Model framework used in the project

3.4.1 GARCH(1,1)

The standard GARCH(1,1) specification models conditional variance as

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2,$$

with

$$\varepsilon_t = r_t - \mu.$$

In the project code, the primary baseline uses a Student- t innovation distribution, so the one-step VaR forecast becomes

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} q_\alpha^{(t)},$$

where $q_\alpha^{(t)}$ is the fitted Student- t lower-tail quantile.

3.4.2 GJR-GARCH

To allow negative shocks to have a different effect on future volatility, the GJR-GARCH model uses

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2.$$

This is useful for equity returns because downside shocks often increase volatility more than upside shocks of the same magnitude.

3.4.3 Historical Simulation

Historical Simulation is a fully nonparametric benchmark. It simply takes the empirical 1% quantile of the most recent W returns,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \text{Quantile}_{\alpha}(r_{t-W+1}, \dots, r_t),$$

with a rolling window of $W = 250$ in the project. It is simple, transparent, and often surprisingly competitive, although it can adapt slowly when volatility changes rapidly.

3.4.4 Filtered Historical Simulation (FHS)

Filtered Historical Simulation combines a parametric volatility model with a nonparametric tail estimate. After fitting a GARCH-type model, the standardized residuals are computed and their empirical α -quantile is used instead of a Gaussian or Student- t quantile:

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_{\alpha, \text{emp}}.$$

The project includes both GARCH-FHS and GJR-FHS, as well as Student- t -based FHS variants in the main runner. These methods are semi-parametric: the conditional scale is modeled parametrically, but the tail is learned from the empirical residual distribution.

3.4.5 Residual bootstrap VaR

The residual bootstrap approach resamples standardized residuals from a fitted GARCH model, simulates one-step-ahead returns, and then takes the empirical 1% quantile of those simulated returns. For one-step forecasting, this is mechanically very close to FHS, which helps explain why the bootstrap and FHS results are similar in the saved outputs.

3.4.6 CAViaR-SAV

CAViaR models the quantile directly instead of passing through a volatility equation. The saved implementation uses the Symmetric Absolute Value (SAV) specification:

$$v_t = \beta_0 + \beta_1 v_{t-1} + \beta_2 |r_{t-1}|,$$

followed by the mapping

$$q_t = -v_t.$$

The resulting one-step VaR forecast is simply the next recursively generated conditional quantile. Parameters are estimated by minimizing the quantile regression check loss

$$L(\beta) = \sum_t \rho_\alpha(r_t - q_t(\beta)),$$

where

$$\rho_\alpha(u) = u(\alpha - I(u < 0)).$$

This approach is theoretically appealing, but the current project implementation is intentionally modest: only the SAV variant is used, optimization is simplified, and no Dynamic Quantile test is included.

3.4.7 GARCHX with VIX

The project also includes a dedicated package-based GARCHX benchmark in R using the `garchx` package. In each rolling window, returns are demeaned, VIX is standardized, and conditional variance is modeled as a GARCHX process with an exogenous volatility regressor. In simple notation,

$$y_t = r_t - \mu,$$

$$\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta x_t,$$

where x_t is the standardized VIX series. The one-step-ahead forecast is converted to VaR via a Gaussian quantile,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \mu + \hat{\sigma}_{t+1|t} \Phi^{-1}(\alpha).$$

This model is best interpreted as a package-based Gaussian-QML benchmark with exogenous volatility information rather than as the final recommended forecasting method.

4 Representative Data Example and Code

4.1 Data example

The data example in this report is the actual project dataset: QQQ daily log returns with VIX as an exogenous volatility proxy. The first few observations in `data/model_data.csv` begin in January 2015, and the rolling backtests in the repository evaluate 848 out-of-sample one-day VaR forecasts.

4.2 Representative code from the project workflow

The following commands summarize the core workflow used to generate the saved project outputs.

```
python Data_Scraping/fetch_data.py --start 2015-01-01 --outdir data
python run_var_project.py --data data/model_data.csv --outdir outputs --alpha 0.01 --initial-train
python run_var_ext.py --data data/model_data.csv --outdir outputs --alpha 0.01 --initial-train
python run_var_riskmetrics.py --data data/model_data.csv --outdir outputs/riskmetrics --alpha 0.01
Rscript GARCHX.r
```

The following project-style Python snippet illustrates the rolling VaR design used throughout the repository.

```
import pandas as pd

# load data
# df contains Date, log_ret, and optionally vix_close

df = pd.read_csv("data/model_data.csv", parse_dates=["Date"])
df = df[["Date", "log_ret", "vix_close"]].dropna().sort_values("Date").reset_index(drop=True)

alpha = 0.01
initial_train = 0.7
n = len(df)
train_end0 = int(n * initial_train)

rows = []
for t in range(train_end0, n):
    train = df.iloc[:t]
    test_row = df.iloc[t]

    r_train = train["log_ret"].to_numpy()
    realized = float(test_row["log_ret"])

    # example placeholder for one fitted model
    var_t = fitted_model_forecast(r_train, alpha=alpha)
    hit_t = int(realized < var_t)
```

```

rows.append({
    "Date": test_row["Date"],
    "VaR": var_t,
    "Return": realized,
    "Violation": hit_t,
})

```

This code structure is important because it enforces a genuine forecast origin and avoids look-ahead bias.

5 Empirical Results

5.1 Summary of saved backtest evidence

The integrated saved backtest results in the repository compare thirteen model variants. Each model is evaluated on the same 848 out-of-sample days at the 1% VaR level. Since the expected number of violations is 8.48, a good model should produce a violation rate near 1% while also avoiding clustered exceptions.

Table 1 summarizes the main results from the repository's saved backtest outputs.

Model	Violations	Violation Rate	Kupiec p-value	Christoffersen p-value	Conditional Coverage p-value
GJR-FHS	8	0.94%	0.8672	0.6961	0.9137
GARCH-FHS	7	0.83%	0.5984	0.7327	0.8212
GARCH-t-FHS	6	0.71%	0.3664	0.7698	0.6371
GJR-t-FHS	6	0.71%	0.3664	0.7698	0.6371
GARCH-t-Bootstrap	6	0.71%	0.3664	0.7698	0.6371
GARCH-Bootstrap	5	0.59%	0.1934	0.8075	0.4166
Historical-Simulation	12	1.42%	0.2528	0.1569	0.1910
GARCH(1,1)	13	1.53%	0.1480	0.1890	0.1482
GJR-GARCH	15	1.77%	0.0424	0.4621	0.0972
GARCHX(VIX)	15	1.77%	0.0424	0.2621	0.0679
[garchx pkg]					
CAViaR-SAV	6	0.71%	0.3664	0.0297	0.0625

Model	Violations	Violation Rate	Kupiec p-value	Christoffersen p-value	Conditional Coverage p-value
RiskMetrics (lambda=0.94)	18	2.12%	0.0043	0.3925	0.0117
GARCH-Gaussian	22	2.59%	0.0001	0.5970	0.0005

Several patterns stand out immediately.

First, the best models are the empirical-tail methods, especially GJR-FHS and GARCH-FHS. They produce violation rates close to the nominal 1% target and also perform well on independence and conditional coverage. This indicates that they are not only getting the total number of violations roughly right, but are also spreading those violations in time in a way that is consistent with a well-calibrated conditional risk model.

Second, purely Gaussian tail modeling performs poorly. The Gaussian GARCH benchmark records 22 violations, far above the expected 8.48, and decisively fails both unconditional and conditional coverage. RiskMetrics also underestimates left-tail risk, though it is not as poor as Gaussian GARCH. These results are consistent with the standard financial observation that daily returns have heavier tails than a Gaussian model allows.

Third, Historical Simulation remains competitive. It is slower-moving and less reactive than the best GARCH-based models, but it produces a reasonably defensible backtest profile and deserves to remain in the final comparison as an interpretable benchmark.

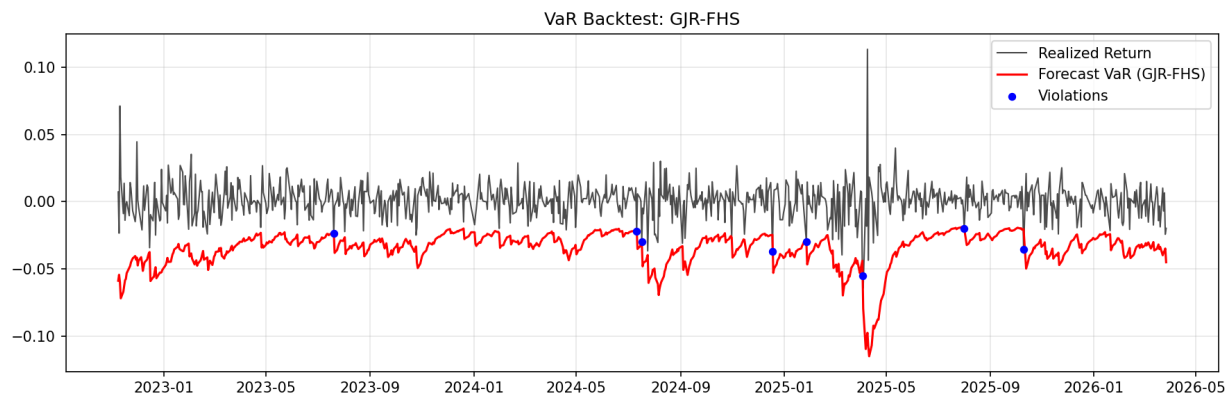
Fourth, the package-based GARCHX model with VIX does not improve on the best models in the saved outputs. Its violation rate is too high, and its backtesting profile is weaker than that of GJR-FHS, GARCH-FHS, and even Historical Simulation. This does not mean VIX is useless, but it does mean that in this implementation VIX-augmented variance forecasting does not automatically translate into better one-day left-tail quantile forecasts.

Finally, CAViaR-SAV is not a complete failure, but it is not among the most trustworthy models either. Its overall violation count is conservative and acceptable, but the Christoffersen test suggests dependence in the hit sequence. That is an important warning sign, especially given the simplified optimization strategy used in the current code.

5.2 Plot-based interpretation

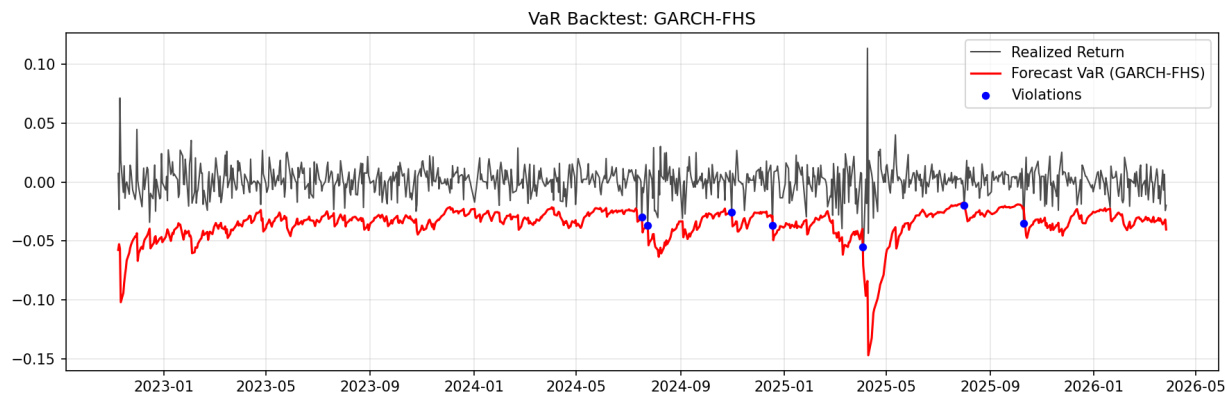
The project repository includes VaR-versus-return plots for each major model. A few of those figures are especially useful for understanding the backtest story.

5.2.1 Best overall model: GJR-FHS



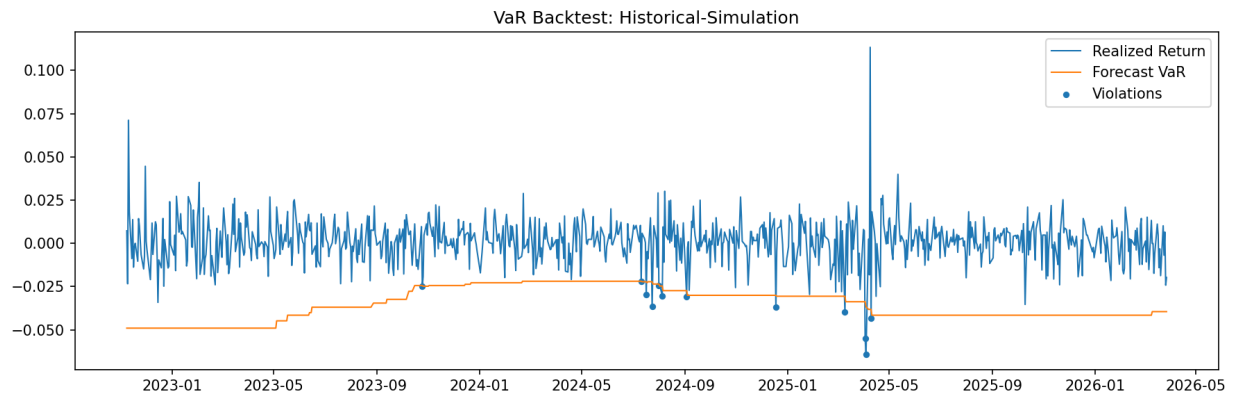
The GJR-FHS forecast line becomes meaningfully more negative during stress periods, which is exactly what a useful left-tail risk model should do. The line is reactive without becoming excessively conservative, which helps explain why it achieves both a near-target violation rate and strong backtest p-values.

5.2.2 Close second: GARCH-FHS



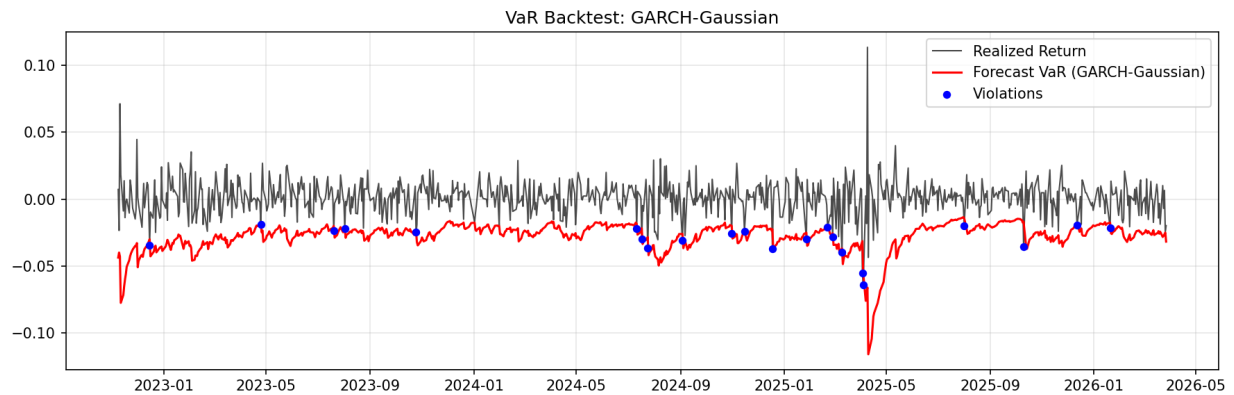
GARCH-FHS behaves similarly, though slightly less asymmetrically than GJR-FHS. The empirical residual tail appears to be doing most of the work, while the GARCH volatility recursion provides the time-varying scale.

5.2.3 Strong nonparametric benchmark: Historical Simulation



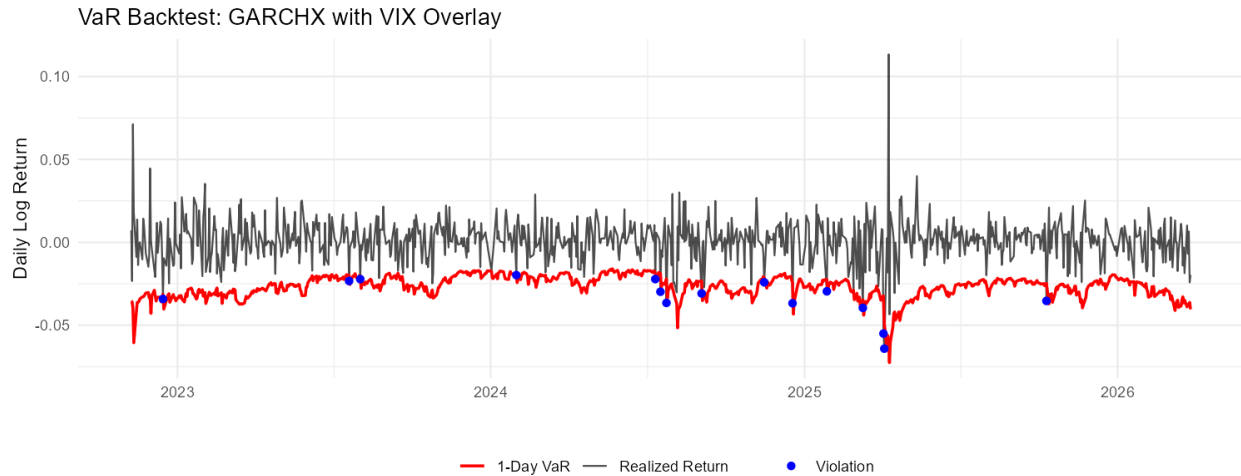
Historical Simulation is visibly more stepwise and slower to adjust than the FHS methods, but it remains a credible benchmark because it still captures downside risk reasonably well without imposing a strong parametric structure.

5.2.4 Weak Gaussian benchmark



The Gaussian GARCH VaR line sits too close to zero during volatile periods. This visual under-reaction is consistent with the backtesting evidence: the model is breached too often because the Gaussian tail is too light for QQQ downside risk.

5.2.5 Exogenous-volatility comparison: package-based GARCHX



The package-based GARCHX benchmark produces a coherent risk series and is useful as a methodological comparison. However, its saved backtesting results remain weaker than the best empirical-tail alternatives, so it is better viewed as a benchmark than as the preferred final model.

6 Discussion

6.1 Why do the empirical-tail models work best?

The main lesson of the project is that better tail calibration matters more than simply changing the volatility recursion. The strongest models all combine time-varying conditional scale with an empirical estimate of the lower-tail innovation distribution. This is exactly what FHS does. It allows volatility to evolve flexibly while avoiding the overly thin tails that often appear in Gaussian VaR.

The GJR-FHS result suggests that adding asymmetry can still help, but only after the tail problem is handled properly. In other words, leverage effects matter, but empirical tail modeling matters more.

6.2 Why does Gaussian VaR fail here?

QQQ returns display volatility clustering and fat tails, especially during market stress. A Gaussian quantile compresses tail risk too aggressively, so one-day VaR forecasts are not negative enough during volatile episodes. The backtest evidence makes this visible both statistically and graphically.

6.3 What is the role of GARCHX with VIX?

The package-based GARCHX model provides a more standard benchmark for testing whether an exogenous market volatility indicator can improve conditional variance forecasts. In principle, VIX should contain forward-looking information about market uncertainty. But in the saved results, that information does not translate into better one-day 1% VaR forecasts than the best FHS models. This is an important empirical reminder: a better volatility regressor does not necessarily imply a better tail-quantile forecast.

6.4 Limitations

Several limitations should be stated clearly.

1. The project workflow now spans multiple runners, so the model comparison is slightly less tidy than in a single integrated script.
2. The package-based GARCHX benchmark uses a Gaussian quantile overlay, so it is still parametric in the tail even though the variance recursion is standard and defensible.
3. The current CAViaR implementation is simplified relative to the original paper: only SAV is implemented, optimization is limited, and no Dynamic Quantile adequacy test is included.
4. The bootstrap implementation resets the random seed internally, which makes the rolling bootstrap effectively deterministic in the saved runs.
5. The conclusions are tied to one ETF, one horizon, and one VaR level. A broader study could compare multiple assets, multiple horizons, and Expected Shortfall as well as VaR.

7 Conclusion

This report converted the project presentation and repository outputs into a written account of one-day-ahead 1% VaR forecasting for QQQ daily log returns. The evidence from the saved rolling backtests is clear: empirical-tail GARCH models are the strongest methods in this project.

The best overall model is GJR-FHS, with GARCH-FHS as a very close second. Student- t empirical-tail variants also perform strongly, and Historical Simulation remains a credible benchmark. In contrast, Gaussian GARCH and RiskMetrics underestimate downside risk, while the package-based GARCHX(VIX) benchmark does not improve upon the best empirical-tail models in the saved outputs. CAViaR-SAV is interesting as a direct-quantile approach, but in its current implementation it remains exploratory rather than definitive.

The broader takeaway is that for one-day-ahead equity VaR forecasting, accurate volatility dynamics are not enough on their own. Tail modeling matters. In this project, the most successful forecasts are those that combine conditional volatility estimation with an empirical description of the innovation tail.

8 References

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